

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns

Perception Software Engineer

Perception Software Engineer with 3+ years' experience building real-time perception and SLAM systems in safety-critical industrial and field environments. Strong ROS2 (rclcpp) and C++17 background across point cloud processing and RTAB-Map SLAM, including deployment on UGV platforms using Velodyne VLP-16. Proven ability to design modular ROS2 pipelines enabling deterministic, repeatable evaluation via rosbag playback, with Linux-based simulation-to-hardware workflows and edge-optimised deployment.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built predictive models for classifying 24/7 oil-rig operations using time-series and CV features, achieving 23% higher F1 and supporting automated risk triage.
- Built an automated sensor-data cleansing pipeline handling 3.6 million records, reducing model rollout time from 3 days to under 24 hours.
- Implemented automated data ingestion, cleaning, training, and validation pipelines for large multimodal sensor datasets using MLflow, DVC, and Optuna, streamlining experimentation and reducing model iteration time by 40%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Collaborated with hardware teams and the client to validate software performance in real-world field tests and enhance system robustness.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Developed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision and fixed-point quantization with structured pruning to meet strict memory and latency budgets for real-time decision-making in industrial environments.
- Developed adversarial-condition training and validation suites simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Built ROS2 node graph with tf2 frame management and sensor message handling to support a modular perception pipeline, enabling deterministic and repeatable evaluation via rosbag playback.
- Tuned SLAM parameters to improve localisation stability across varied terrains, achieving consistent behaviour in feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

Jan 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration within a modular perception architecture.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Denoising, Eigen, LiDAR SLAM, NumPy, Open3D, OpenCV, Point Cloud Library, ROS2 image_transport, SAM, Scan Matching, Scikit-Image, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Git, Intel VPUs, Linux, Microsoft Azure, MLFlow, NVIDIA Jetson
- **Programming & Development:** Bash, C++17, CMake, colcon, Django, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, Qt, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)

- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT